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Unit - V Pharmacology - 2

• pharmacology of drugs acting on Endocrine system

- Androgens and Anabolic steroids

- Estrogen, progesterone and Oral contraceptives

sex hormones.

- Drugs acting on uterus.

Bioassay

- principles and applications of bioassay

- Types of bioassay

- Bio assay of Insulin, Oxytocin, vasopressin, ACTH, ~~dehydro~~ d-tubocurarine, digitalis, Histamine and 5-HT

ImpQues

① Write a details note on Oral contraceptives.

② Write a note on drugs acting on uterus.

③ Write a short note on:-

• Androgens and Anabolic steroids

• Estrogens and Progesterone

④ Define and classify Bioassay. Write the bioassay of _____ drugs are ??

• Insulin • d-tubocurarine

• Oxytocin • Digitalis

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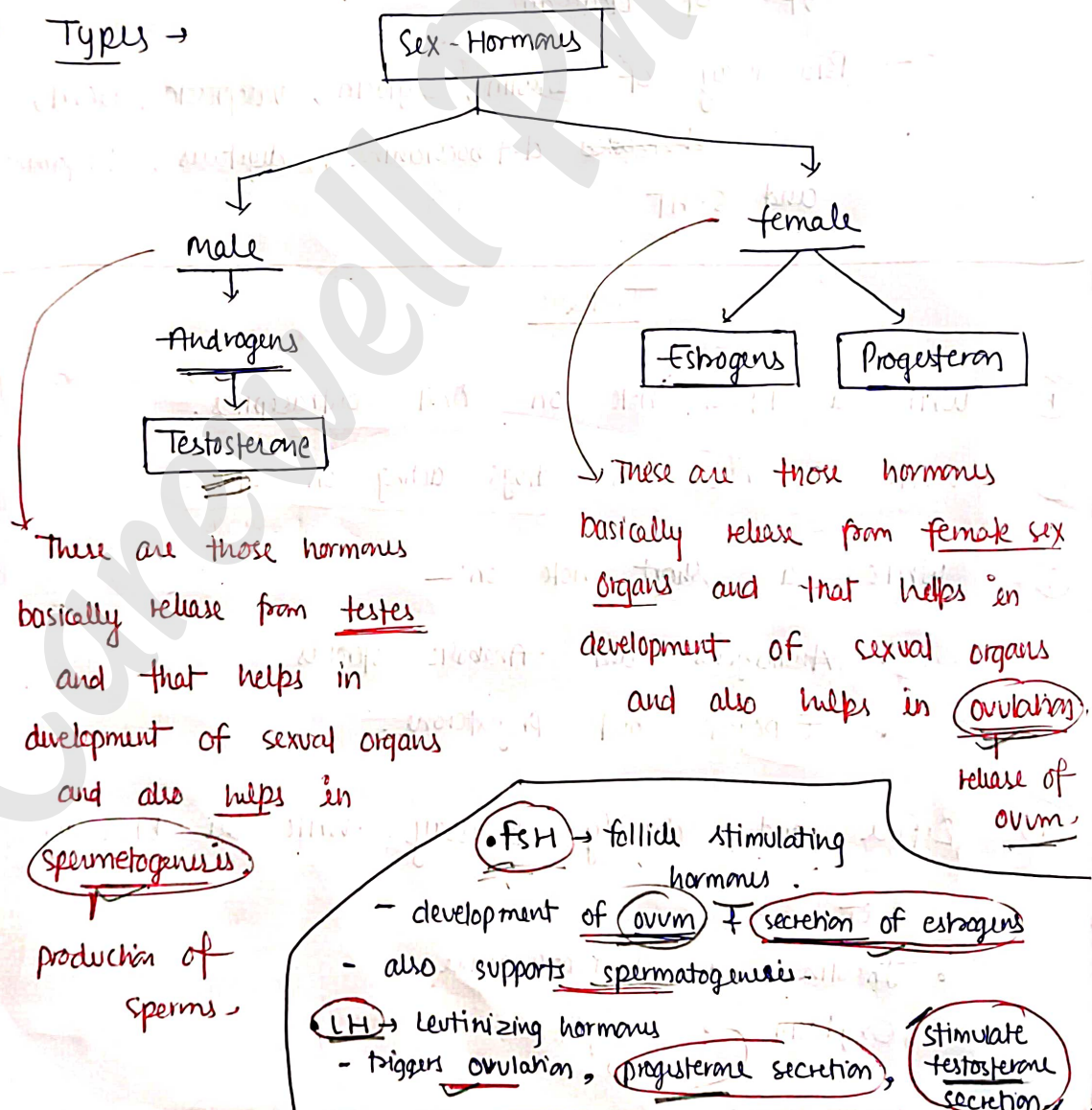
Basic knowledge →

• Sex hormones →

These are those hormones which produced and released in human body and they play an important role in sexual development & reproduction.

other names → Sex steroids, Gonadocorticoids, Gonadal steroids

• Types →



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① Write a short note on

- Androgen and Anabolic steroids

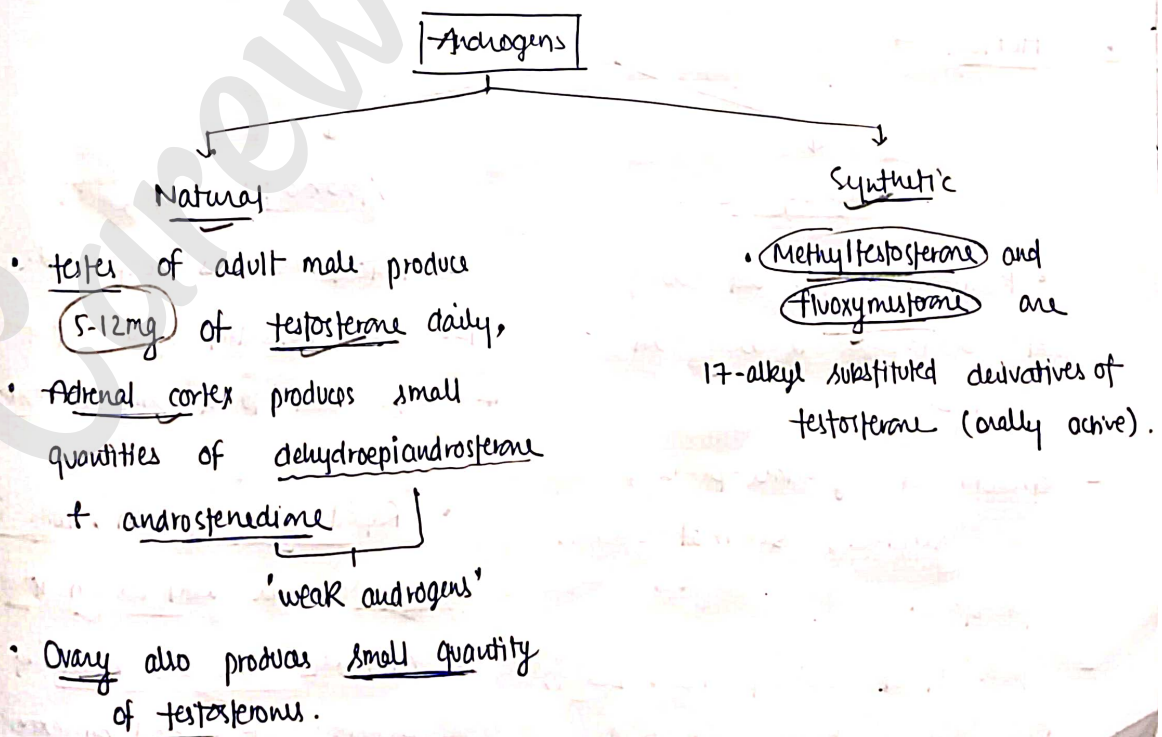
→ Androgens → It is male sex hormones.

- Testosterone is the main natural androgen. It is mainly synthesized by the interstitial cells of the testis, and in smaller amounts by the ovaries and adrenal cortex.
sex steroids.

- These substances cause development of secondary sex characters in the men, including facial and body hair growth and voice change.

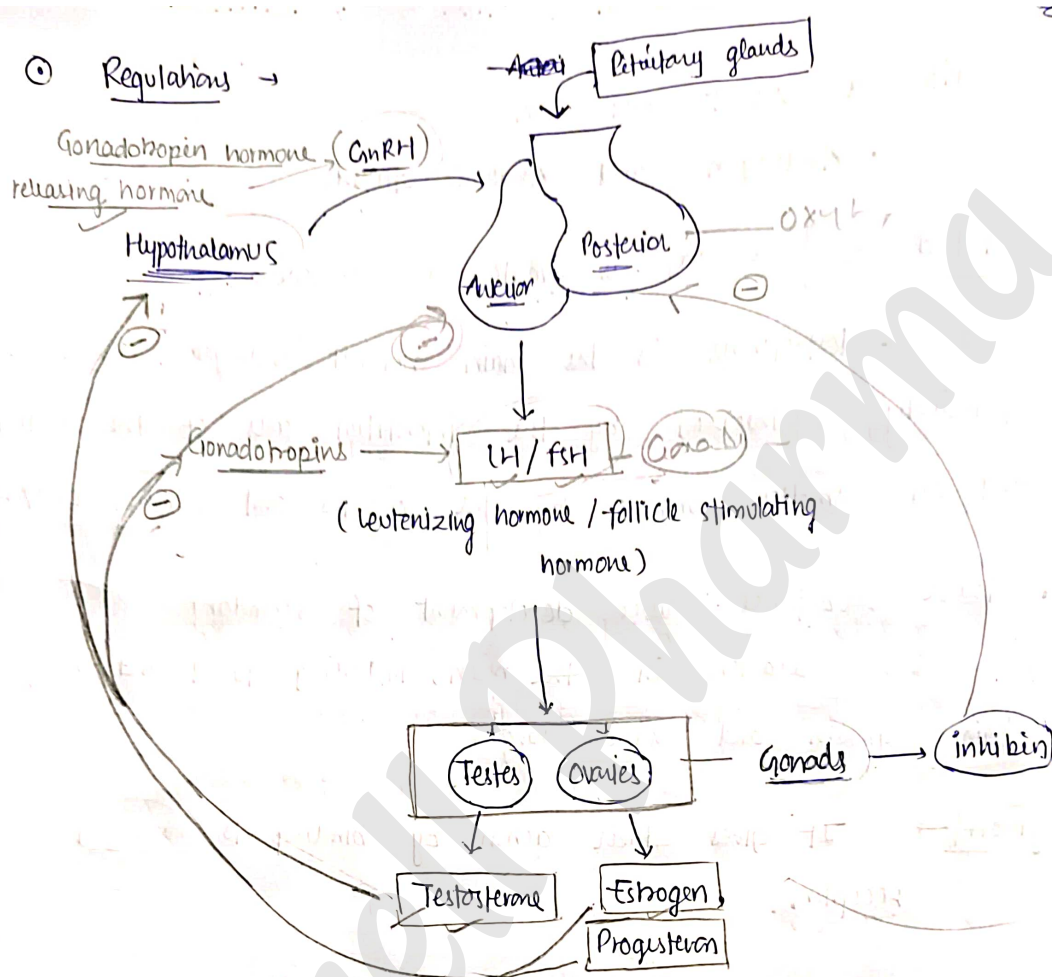
Mechanism → It gives their action by binding to androgen receptor.

② Androgens are may be natural & synthetic



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② Action →

- | | |
|---|--|
| <p><u>Androgenic</u></p> <ul style="list-style-type: none"> • Responsible for all changes that occurs in a boy in <u>puberty</u>. - growth of <u>genitals</u>-penis, <u>scrotum</u>, seminal vesicles, prostate. - growth of <u>hair</u>-pubic, <u>beard</u>, <u>moustache</u>, <u>body hair</u> and | <p><u>Anabolic</u> enter into puberty</p> <ul style="list-style-type: none"> • testosterone is responsible for the <u>pubertal spurt</u> of growth in boys and to a <u>smaller extent</u> in girl. • <u>Rapid bone growth</u>, both in <u>thickness</u> as well as in <u>length</u>. • <u>promote muscle building</u>, especially if <u>aided</u> by <u>exercise</u>. |
|---|--|

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male pattern of its distribution.

- larynx grows & voice deepens.
- etc etc -

- Appetite is improved and a sense of well-being prevails.

• pharmacokinetics →

- Usually ~~given~~ ~~intrac~~ ~~ly~~ injected.
- bind with plasma protein globulin
- plasma $\frac{1}{2}$ life → 10-20 minutes.

Anabolic steroids → (Anabolic androgenic steroids).

- They are synthetic androgens with supposedly

higher anabolic and lower androgenic activity.

- Rapid bone growth
- promote muscle building.
- Sexual growth

- These are those chemical substance which increases the size of the muscles and helps to grow the body

— sometimes taken illegally by people who play sports.

(eg) Nandrolone, Methandienone, etc..

• Antiandrogens → Those drugs which antagonise the action of androgens. used to decrease the excessive

sexual desire, (eg) flutamide, cyproterone etc..

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- Estrogens →
also known as oestrogen.
- It is a primary female sex hormone.
- It is responsible for the development and regulation of the female reproductive system and secondary sexual characteristics.
- Sex-steroids are involved in the regulation of the cyclic changes expressed in menstrual cycle and are important in pregnancy.
- Synthesized by ovary and placenta, and in small amounts by the testis and adrenal cortex.
- There are three major endogenous estrogens in females:— Estroene (oestroene), Estradiol (oestradiol) and estriol (oestriol).
- Estradiol is the most potent and it is the principal estrogen secreted by ovary.
- Synthetic → Ethinylestradiol, Mestranol etc.
- Regulation → already discussed.

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- main function →

- pubertal change in females
- growth of uterus, fallopian tubes and vagina
 ↓
 reproductive organ.
- vaginal epithelium gets thickened
- responsible for proliferation of endometrium in the preovulatory phase
- estrogens increase the coagulability of blood, and increase the risk of thromboembolism.

- clinical uses →

- primary ~~ovary~~ ovarian failure (eg. turner's syndrome)
- secondary ovarian failure for flushing, vaginal dryness and to preserve bone mass.
- contraception — in combination pills
- prostate and breast cancer.

• Progesterone →

- It is secreted by the corpus luteum ^(ovary) in the second part of the menstrual cycle, and by the placenta during pregnancy.
- Small amounts are also secreted by testis and adrenal cortex.
- These are substances which convert the estrogen. ~~primarily~~

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— main functions →

- It brings about secretory changes in the estrogen primed endometrium.
- It ^{It cause} thinned out mucus in the cervix to become thick again which prevent sperm from passing through.

- It induces pregnancy.

- It cause proliferation in the mammary glands.

— It also inhibit secretions of pituitary gonadotropins which prevent ovulation.

② Write a details note on Oral contraceptives

→ Oral contraceptives →

Also known as "Birth control Pills".

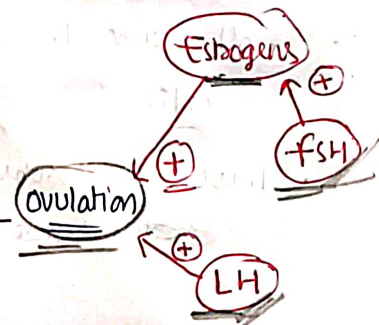
- These are those drugs which are used to prevent pregnancy. prescription medications

- These medicines contain hormones like estrogen and progestin (synthetic but similar as progesterone)

- Role of these hormone in pregnancy [normal mechanism] →

— The ovaries produce the egg cells called ovum (ova) then it transported into fallopian tube.

| release

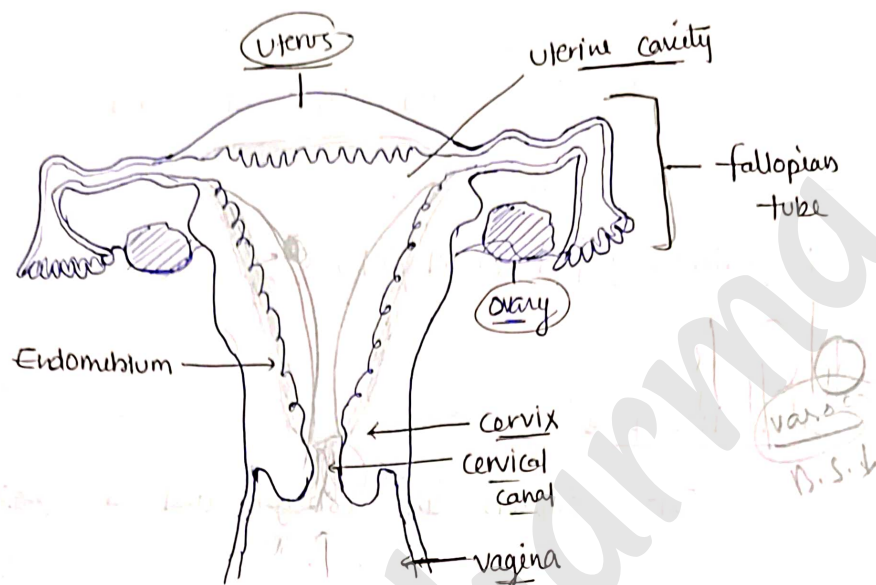


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- Now this egg fertilized by sperm in fallopian tube. *
- Now, this egg moves to uterus, where the uterine lining has thickened and prepare to receive a fertilized egg. (+)



- Progesterone
 - Rising level of progesterone that cause glands in lining to release fluids that feeds the egg. (fertilized egg). endometrium
 - It also causes the thinned out mucus in the cervix to become thick again which prevent sperm from passing through.
- * If an egg hasn't been fertilized the level of both estrogen and progesterone begin to fall which cause menstruation.

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- Oral contraceptives →

there are two main types of oral contraceptives →

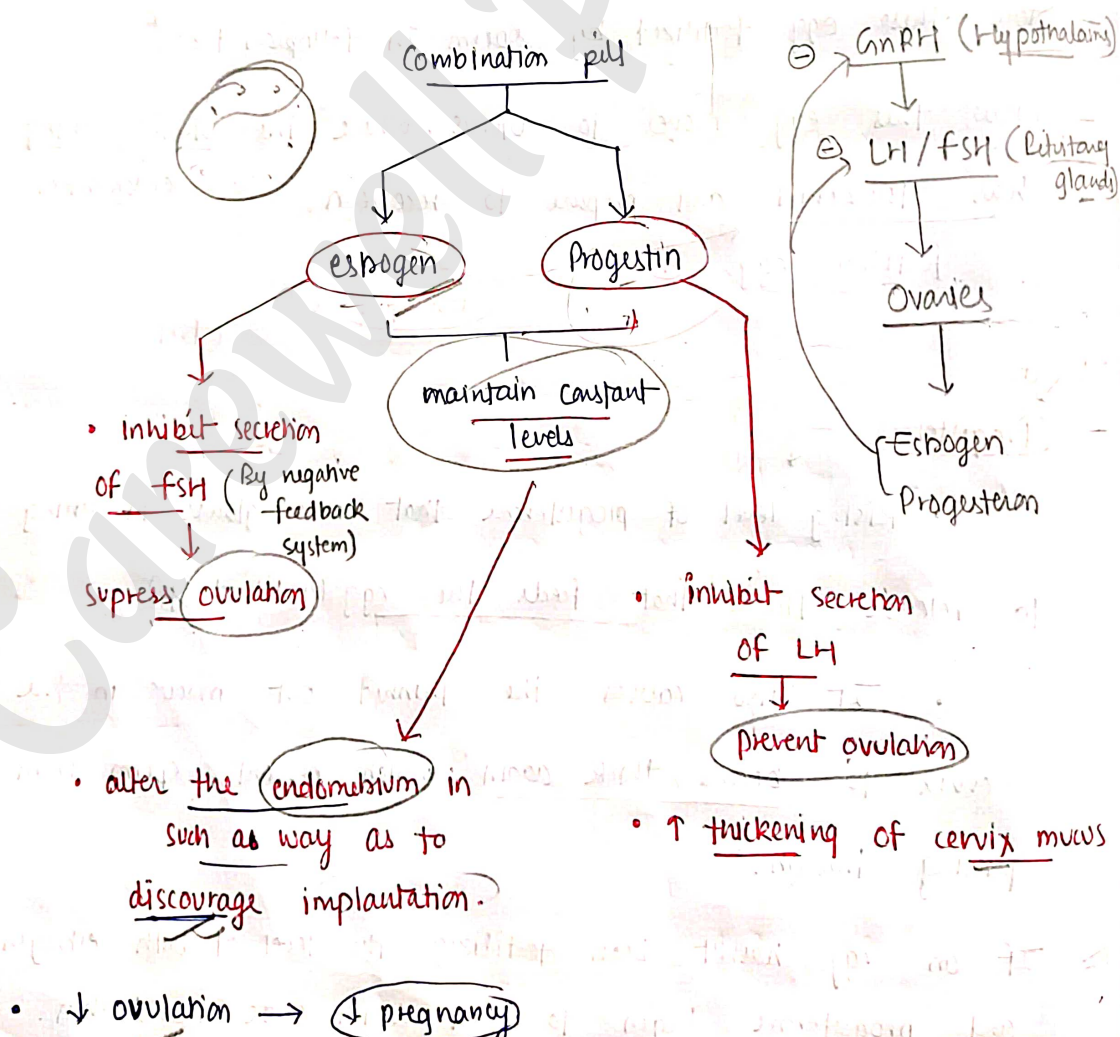
- i) Combination pills
- ii) Progestin-only pill (mini pill)

- i) Combination pills →

It contains an estrogen and a progestin

Synthetic version of progesterone.

MOA →



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• this is most popular and most efficacious method.

eg) Ethinylestadiol & Norgestrel etc.

Adve → weigh gain, nausea, vomiting etc.

① Course → Day wise → < 1 active pills that contain hormones.

ii) Minipill (progestin only pill) →

They contain only progestin.

MOA → prevent ovulation & thickening of cervical mucus
prevent pregnancy

- their efficacy is lower than combination pills.

③ Write a note on drugs acting on uterus.

-
- uterine muscle contracts rhythmically both in vitro and in vivo, the contraction originating in the muscle itself.
 - myometrial cells in the fundus act as pacemaker and give rise to conducted action potentials.
 - Now, these activity of pacemaker is regulated by the sex hormones.

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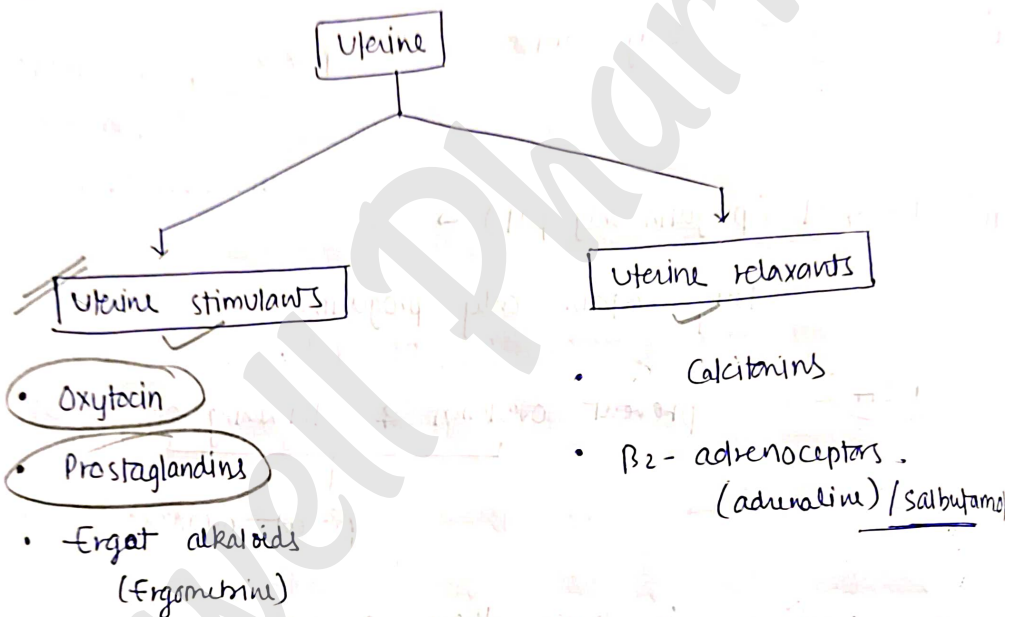
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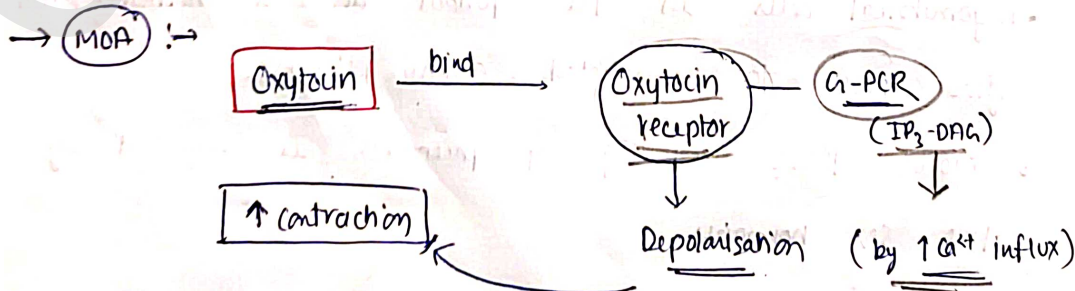
- uterine contraction (contracts spontaneously)
 ↓
 strong during menstruation.
- fully coordinated during parturition. — giving birth

Classification →



① Oxytocin →
 It is a ~~hormone~~ ^{hormone} secreted by the posterior pituitary along with vasopressin (ADH).

- It increase the contraction of uterine muscles.



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- ~~Also~~ The number of oxytocin receptors increases markedly during later part of pregnancy

↓
[contraction] → Help in child birth

Uses :- Action →

- It is used to induce or augment labour when the uterine muscle is not functioning adequately
- Also used to treat postpartum haemorrhage

↓
after immediately child birth.
(excessive blood loss)

- Breast → increase the secretion of milk from the mammary glands.

Adverse effects :-

In high dose →

- cause sustained contraction that interfere with blood flow through the placenta and cause fetal death.

Pharmacokinetics →

- given by I.V or I.M but mostly Intravenous Infusion.
- Inactivated in liver & kidney.

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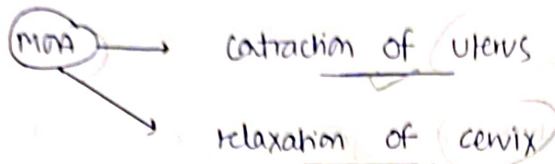
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② Prostaglandins →

PGE₂, PGF_{2α} and 15-methyl PGF_{2α} are potent endogenous uterine stimulants.



③ Ergot Alkaloids →

It contracts the human uterus.

- On a contracted uterus (the normal state following delivery), ergometrin has relatively little effects.
- However, if the uterus is inappropriately relaxed, ergometrine initiates strong contraction, thus reducing bleeding from the placental bed.

Uterus → same as oxytocin.

• Uterine relaxants →

- Selectively β₂-adrenoceptor agonists, such as ritodrine or salbutamol, inhibit oxytocin-induced contraction of the pregnant uterus.

— used to prevent premature labour (delay delivery upto 48 hrs).

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BIOASSAY

• (Bio) + (Assay)
in living being test

→ It is the process of determining the potency of the drug by using suitable biological system like, animals, tissue, microbes etc..

- Determine relative potency of a substance (drug) by comparing its biological activity with that of a reference standards.



- Also it is used to standardize the preparation so that each preparation contain specified pharmacology activity.

• Principle →

The basic principle of bioassay is to compare the test substance with the international standard preparations of the same and to find out ~~the~~ how much test substance is required to produce the same biological effects, as produced by the standard.

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Applications →

- to determine the potency of a drug
- to determine the dose (quantity) of a drug required to produce a therapeutic or toxic response, eg. Calculation of ED₅₀, LD₅₀ etc. -
 - effective dose
 - lethal dose
- Also helps in the development of new drugs.

Types of bioassay →

There are mainly two types:— (Acc. to their responses).

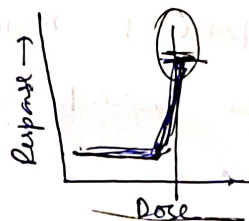
- Quantal bioassay
- Graded bioassay

i) Quantal bioassay →

Also known as "All or None (No)" & "End point".

- Response is in the form of "All or none", i.e. either maximum response or no response, there are no intermediate responses.

eg. Digitalis → Cardiac arrest
in guinea pig / pigeon
d-tubocurarine → by rabbit head
drop method



- comparison of threshold response & comparison of ED₅₀ or LD₅₀.

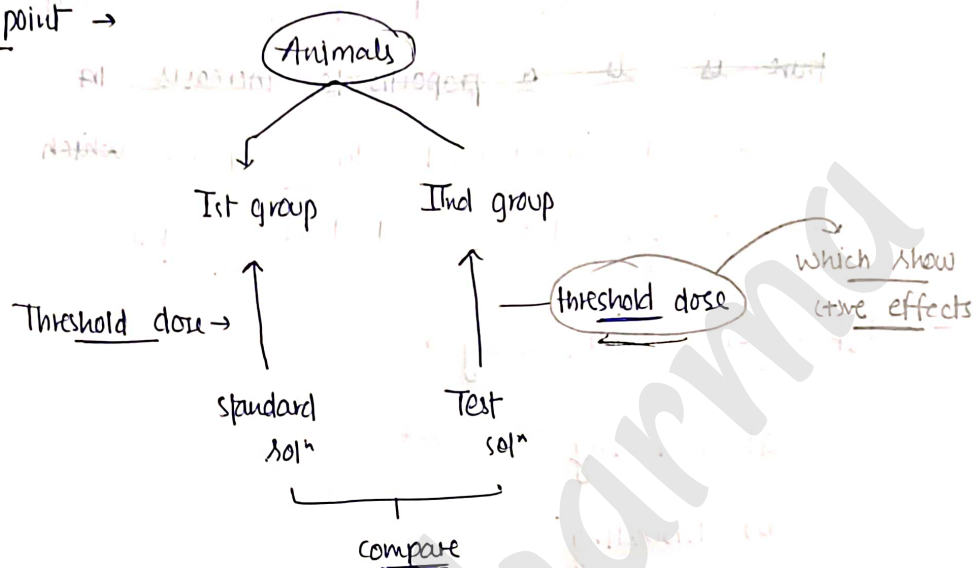
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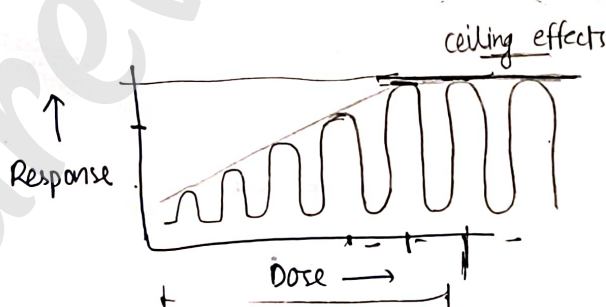
• End point →



– Concentration of unknown $\Rightarrow \frac{\text{Threshold dose of standard}}{\text{Threshold dose of test}} \times \text{conc. of standard.}$

ii) Graded bioassay →

It means the response is proportional to the dose and response may lie between no response and the maximum response.



- The graded dose response curve is obtained by plotting a graph with dose on the x-axis and response on the y-axis.
- It is usually sigmoid in shape.

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- These responses are based on the observations that

there is an increase in responses with increase in the concentration of dose.

- Types of graded assay →

a) matching

b) Bracketing

c) Interpolation

d) Multiple points bioassay

three point

four point

six point

eight point

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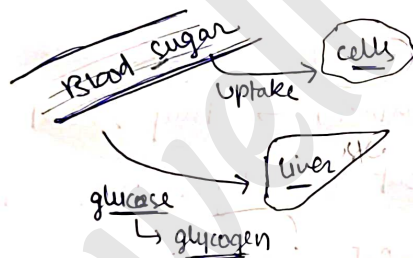
① Bioassay of insulin →

Insulin →

- It is a hormone made by pancreatic β -cells.
- It is synthesized as pro-insulin in endoplasmic reticulum and is processed to the biologically active form inside the secretory granules.

function →

- It helps to control the blood sugar level by converting glucose into glycogen and also increase the glucose uptake.



- Also write def. of bioassay + Principle (std, test)

Experimental models →

- Rabbit method
- mouse method

preparation of samples →

- Standard → It is pure, dry and crystalline insulin.

1 unit → 0.04082 mg (specified by ministry of health, Govt. of India & also equivalent to international unit).

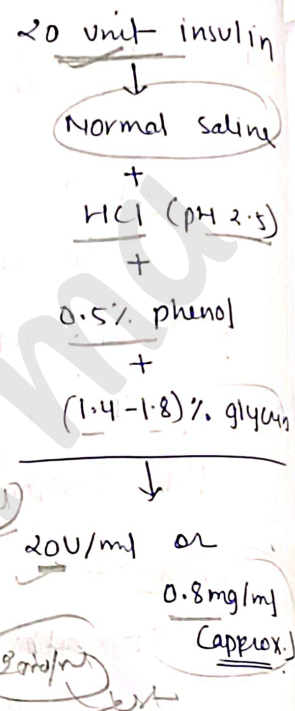
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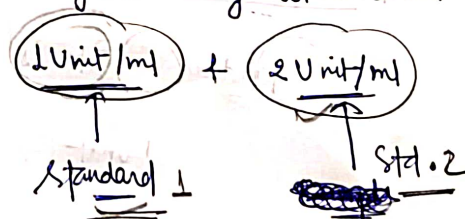
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- Accurately weight 20 units of insulin and dissolve it in normal saline.
- Acidify it with HCl to pH 2.5
- Add 0.5% phenol as preservatives
- Add 1.4% - 1.8% glycerin.
- final volume should contain 200/ml.
- Store the solⁿ in a cool place and use it within six months
- Test → same as standard.



① Rabbit method →

- Selection of Rabbits (animals) → They should be healthy, weight about (2-3 kg). All criteria should be follow acc. to Govt. such as CPCSEA etc.
- for perform test → Rabbits should be maintained on uniform diet but are fasted for 18hrs before assay (test).
- Standard and sample dilutions → these are freshly prepared by diluting with normal NaCl solⁿ so as to contain
 - similarly, test sample solⁿ are solutions are also prepared.



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• principle → The potency of a test sample is estimated by comparing the hypoglycemic effects of the samples with that of std. preparation of insulin.

• procedure →

Animals are divided into 4 groups of 3 rabbits each.

- The rabbits are then put into animal holder.

- They should ~~behave~~ ~~hand~~ handled with care to avoid excitement.

- It perform in two parts:— first part + second part

① first part →

- A sample of blood is taken from the marginal ear vein of each rabbit

- Presence of reducing sugar is estimated per 100ml of blood by a suitable chemical method

- This concⁿ is called "initial blood sugar level".

• Then four groups of rabbits are then given subcutaneous (sc) injection of insulin as follow

Rabbit	Group	Sample	
3	I	Std-1 - 1U/ml	} 0 std
3	II	Std-2 - 2U/ml	
3	III	Test-1 - 1U/ml	} 0 test
3	IV	Test-2 - 2U/ml	

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from each rabbit, a sample of blood is withdrawn upto 5 hrs at the interval of 1hr each.

- Now Blood sugar is determined again & this is called "final blood sugar level".

① Second part →

- After one week choose same animals
- again fasted for 18 hrs and initial blood sugar is determined.

Now, the main change is to reversed the group.

- means - those animals which received the standard are given the test and those which received the test are now given the standard.
- Also - those animals which received the less dose of standard are given the higher dose of the test sample and vice-versa.
- This test is known as "Twin cross over test".

<u>First part</u>		<u>Second part</u> → check sugar level same as first part.	
<u>Grp</u>	<u>Sample</u>	<u>Grp</u>	<u>Sample</u>
I	Std-1 (1)	I	Test-2 (2)
II	Std-2 (2)	II	Test-1
III	Test-1 (1)	III	Std-2
IV	Test-2 (2)	IV	Std-1

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mean percentage of decrease in blood sugar of first and second part is calculated.

② mouse method →

- Mice show characteristic convulsions after s.c. injection of insulin at elevated temperatures.
- The percentage convulsions produced by the test and standard preparations are compared.

procedure → (Animals Selection)

- minimum 100 mice weighing between 18-22gms. of the same strain are used.
- They should be maintain on constant diet.
- They should be fasted 18 hrs prior to the experiment

Sample preparation →

- Dilutions are prepared with sterile solⁿ, so as to contain 0.064 units/ml (std. I) and 0.096 units/ml (std. II).
- Similarly, test sample solⁿ are prepared.
- Mice are divided into four groups each containing 25 unit.

- Insulin is injected s.c. as follow:-

Grp-I	Sample
I	<u>Std. 1</u>
II	<u>Std. 2</u>
III	Test. 1
IV	Test. 2

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- mice are put in an air incubator at 33°C and observed for one and a half hour.
 - The mice which convulse or die are taken out of the incubator and observed.
 - Percentage convulsions produced by test samples are compared with those of the standard sample.
- # Convulsive mice may be saved by an inj. of 0.5 ml of 5% dextrose solⁿ.
- * Those animals which survive may be used again for another experiment after an interval of one week.

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- d-Tubocurarine →

It causes the relaxation of skeletal muscles.

- Bioassay by using Rabbit-head-drop method.

- Principle →

- It is injected into a marginal vein of a rabbit's ear till the rabbit's neck muscles are relaxed such as that the animal cannot hold its head up.

- The total amount of test sample required to produce the endpoint is compared with the total amount of the standard sample required to produce similar effect.

- Selection of Rabbits →

- weigh - 2kg
- should be free from disease
- choose from a healthy colony
- should be accustomed with the experimental procedure

- procedure →

- Rabbit is placed in a holder with its head protruding outside. The head should be freely movable.

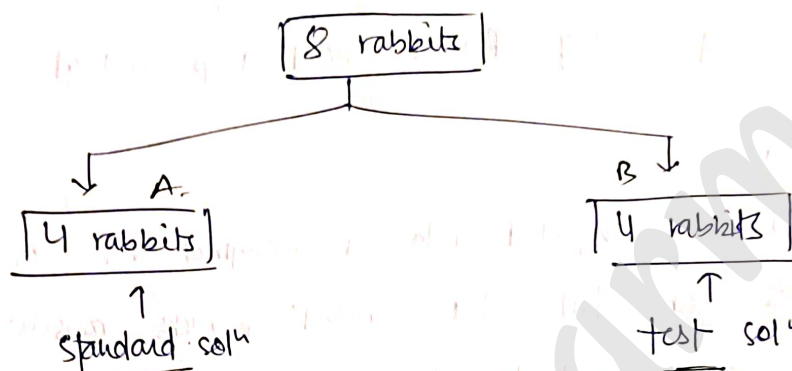
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- min. 8 rabbits are used. They are divided into two groups, each containing 4 rabbits



injected at constant speed

by infusion apparatus through the marginal vein

- speed 0.4 ml/min (time taken about 10 min)

Dose 0.012 % w/v in saline

- infusion is continued till the rabbit will not be in a position to hold its head erect.
- Again cross-over test is carried out to minimize biological errors due to animal variation, those rabbits which received the standard sample on the first day will be given test sample on the second day of experiment or vice versa
- Mean dose which produces head drop of the test sample is compared with the mean dose of standard preparation.

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During the experiment there is possibilities of respiratory embarrassment which is treated by injecting neostigmine methyl sulphate (0.05 mg) and atropine sulphate immediately through the marginal ear vein.

• Bioassay of Digitalis

- It helps to increase the force of contraction of heart

principle →

Potency of test sample is compared with that of standard preparation by determining the action on the cardiac muscle.

Sample →

the standard preparation is a mixture of dried and powdered digitalis leaves (1U = 76 mg).

- Exact amount of the powder is extracted with dehydrated alcohol in soxhlet apparatus for six hours.
 - final extract should contain 10 ml (5 ml alcohol + 5 ml water)
- per 10 g of digitalis powder.

- should be stored in b/w 5°C - 5°C

Animals →

Guinea-pig method (End point)

- Reagan method

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1) Guinea - pig method

Sample + 1g digitalis powder is diluted to 80 ml.

Guinea - pig

anaesthetized

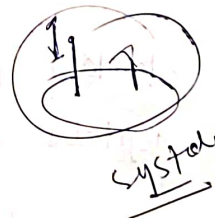
↓
dissected on the
operation table.

↓
Jugular vein is traced out
by removing adhering tissue
and cannulated by means of

venous cannula

↓
pin is inserted in the
heart (in the apex of heart)

↓
we can observe heart
beat by up & down movements
of the pin.



↓
Injected is continued through the venous cannula
until the heart is arrested in systole.

(the amount of extract required to produce this

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effect is taken as the lethal dose of the extract).

- Another set of 19 animals of the same species are used for this experiment and the average lethal dose is determined.

* not necessary to determine the lethal dose of the standard during each time of experiment (but checked).

* finally potency is calculated in relation to standard by dividing the average lethal dose of the sample of test and expressed as units per gram.

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BIOASSAY OF OXYTOCIN:

- Oxytocin is a peptide hormones and Neuropeptide. Oxytocin is normally produced by the paraventricular nucleus of the hypothalamus and released by the posterior pituitary.
- Oxytocin is a natural hormone that causes the uterus to contract. Oxytocin is used to induce labor or strengthen labor contractions during childbirth, and to control bleeding after childbirth.
- Oxytocin is also used to stimulate uterine contractions in a woman with an incomplete or threatened miscarriage.
- Different bioassay methods used for oxytocin are
 - By contraction of the rat uterus

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- By depression of the blood pressure in chicken
- By measurement of milk ejection pressure in a lactating rat etc.

Principle:

- The potency of oxytocin injection is determined by comparing its activity with that of the standard preparation of oxytocin under the conditions of the following method of assay.

Standard preparation:

- The standard preparation is consisting of a freeze dried preparation of oxytocin with human albumin and citric acid (supplied in ampoules containing 12.5 units) or any other suitable preparation, the potency of which had been determined in relation to the International standard.
- The Unit is the specific oxytocin activity corresponding to that yielded by 0.0005 g of the Standard preparation.
- The standard preparation is as per the 4th international standard for Oxytocin, established in 1978.

Experimental Methods:

1) By contraction of the rat uterus:

- Use female rats weighing between 120 and 200 g. Immediately before the assay confirm the rat is in oestrous or preoestrus by vaginal smear.
- Inject 100 microgram of oestradiol benzoate intramuscularly 18-24 hours before the assay. Kill the rat and suspend one horn of the uterus in a bath containing a solution of the following composition:

Composition	(% w/v)
Sodium chloride	0.662
Potassium chloride	0.045
Calcium chloride	0.007
Sodium bicarbonate	0.256
Disodium hydrogen phosphate	0.029
Sodium dihydrogen phosphate	0.003
Magnesium chloride	0.010
Dextrose	0.050

- Maintain the bath at a temperature of 32° C so that spontaneous contraction of the uterus are abolished and preparation maintain its sensitivity.
- Oxygenate the solution with a mixture of 95% of oxygen and 5% of carbon dioxide.

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- Record the contractions of the muscles produced by the addition to the bath of two doses of the Standard Preparation suitably diluted with the above solution. The doses should be such as to produce clearly discriminated, submaximal contractions. The required doses normally lie between 10 and 50 micro Units per ml of bath liquid.
- When maximal contraction has reached replace the bath liquid by a fresh solution.
- The doses should be added at regular intervals of 3 to 5 minutes depending upon the rate of recovery of the muscle.
- Dilute the preparation being examined so as to obtain the responses on the addition of two doses similar to those obtained with the Standard preparation.
- The ratio between the two doses of the preparation being examined should be the same as that between the two doses of the Standard Preparation and this ratio should be kept constant throughout the assay.
- The two doses of Standard preparation and the two doses of the test preparation should be given according to a randomized order or Latin square design and at least six responses to each should be recorded.
- Measure all the responses and calculate the result of the assay by statistical methods.

2) By depression of the blood pressure in chicken:

- Anaesthetize a young healthy adult cockerel weighing 1.2 to 2.3 kg with an anesthetic that will maintain a prolonged and constant high blood pressure.
- Expose the gluteus primus muscle in one thigh and cut and retract it to reveal the popliteal artery and crural vein.
- Cannulate the popliteal artery and record the blood pressure. Cannulate the crural or brachial vein.
- Prepare standard solution with saline so that the volume to be injected is between 0.1-0.5 ml.
- Inject 2 doses of standard solution into cannulated vein and record blood pressure. The doses should be such that as to produce clearly discriminated, precipitous, submaximal decreases in B.P.
- The required doses normally lie between 20 and 100 milli units.
- The interval between injections should be constant and lie between 3-10 minutes depending on the rate at which B.P. returns to normal.
- Dilute test sample before use with saline solution so as to obtain responses similar to those obtained with the standard preparation.

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- The ratio between the two doses of the test preparation being examined should be same as that between the two doses of the standard preparation and the ratio should be kept constant throughout the assay.
- The two doses of Standard preparation and the two doses of the test preparation should be given according to a randomized order or Latin square design and at least six responses to each should be recorded.
- If animal rapidly become insensitive due to repeated injections of the solutions another animal must be used. Measure all responses and calculate the result by standard statistical methods.

3) By measurement of milk-ejection pressure in a lactating rat:

- Select a lactating rat, in the third to twenty-first day after parturition and weighing about 300 g separate it from the litter and 30 to 60 minutes later anaesthetise (for example, by the intraperitoneal injection of a solution of Pentobarbitone Sodium).
- Tie the rat to an operating table, maintained at 37°C, by its hind legs leaving the front legs free. Cannulate the trachea with a short polyethylene tube of internal diameter about 2.5 mm in such a manner so as to ensure a free airway; apply artificial respiration only if necessary.
- Cannulate an external jugular or femoral vein with a polyethylene tube of internal diameter about 0.4 mm which is filled with saline solution and closed with a pin.
- Shave the skin surrounding the inguinal and abdominal teats and excise the tip of one teat, preferably the lower inguinal teat. Insert a polyethylene tube of internal diameter about 0.3 mm and external diameter about 0.6 mm, to a depth sufficient to obtain appropriate measurement of pressure (3 to 10 mm depth), into the primary teat duct which opens onto the cut surface and tie firmly in place with a ligature.
- Connect this cannula with a suitable strain gauge transducer (such as that used for recording arterial blood pressure in the rat) and fill the whole system with a 3.8% w/v solution of sodium citrate or saline solution containing 50 Units of heparin sodium per ml to prevent clotting of milk.
- After cannulation, inject a small volume (0.05 to 0.2 ml) of this solution into the teat duct through the transducer to clear the milk from the tip of the cannula. (This procedure may be repeated during the assay should obstruction arise from milk ejected into the cannula).

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- Clamp the strain gauge so that a slight tension is applied to the teat and its natural alignment is preserved and connect the gauge to a potentiometric recorder adjusted to give full-scale deflection for an increase in milk-ejection pressure of about 5.3 kPa.
- Inject all solutions through the venous cannula using a 1-ml syringe graduated in 0.01 ml and wash them in with 0.2 ml of saline solution.
- Prepare a solution of the Standard Preparation and a solution of the preparation being examined in saline so that the volume to be injected is between 0.1 ml and 0.4 ml.
- Choose two doses of the Standard Preparation such that the increase in milk-ejection pressure is about 1.35 kPa for the lower dose and about 2.7 kPa for the higher dose.
- As an initial approximation, a lower dose of between 0.1 and 0.4 milliUnit and an upper dose of 1.5 to 2 times this amount may be tried.
- Choose two doses of the preparation being examined with the same inter-dose ratio, matching the effects of the doses of the Standard Preparation as closely as possible.
- Inject the four doses (two doses of the Standard Preparation and two doses of the preparation being examined) at intervals of 3 to 5 minutes.
- The two doses of Standard Preparation and the two doses of the preparation being examined should be given according to a randomised block or a Latin square design and at least four responses to each should be recorded. Measure all the responses and calculate the result of the assay by standard statistical methods.

BIOASSAY OF VASOPRESSIN:

Biological Assay for Vasopressor Activity

Principle:

- The vasopressor activity is estimated by comparing the activity of the preparation being examined with that of the Standard Preparation of arginine vasopressin under the conditions of a suitable method of assay.

Standard Preparation:

- The Standard Preparation is the 1st International Standard for Arginine vasopressin, established in 1978, consisting of freeze-dried synthetic arginine vasopressin peptide acetate with human albumin and citric acid (supplied in ampoules containing 8.20 Units), or another suitable preparation the potency of which has been determined in relation to that of the International Standard.

Suggested Method:

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- Inject slowly into the tail vein of a male albino rat weighing about 300 g a solution of a suitable alpha-adrenoceptor blocking agent. For example 10 ml per kg of body weight of a solution prepared by dissolving 5 mg of phenoxybenzamine hydrochloride in 0.1 ml of ethanol (95%), adding 0.05 ml of 1M hydrochloric acid and diluting to 5 ml with saline solution.
- After 18 hours, anaesthetise the rat with an anaesthetic that will maintain a prolonged and uniform blood pressure. After 45 to 60 minutes, tie the rat on its back to the operating table by its hind legs.
- Cannulate the trachea with a short polyethylene tube of external diameter about 2.5 mm and dissect a carotid artery ready for cannulation. Then cannulate the femoral vein close to the inguinal ligament.
- Retract the abdominal muscles to expose the inguinal ligament. Retract the superficial pudendal vein to one side and dissect the femoral vein towards the inguinal ligament from the corresponding artery.
- When dissecting, a deep branch reaching the femoral vein must be found and tied off to prevent bleeding during cannulation. Tie a short polyethylene cannula of external diameter about 1 mm into the femoral vein by two ligatures and join by a short piece of flexible tubing to a 1-ml burette with an attached thistle funnel containing saline solution at about 37.
- Firmly fix a wet absorbent cotton swab to the thigh so as to cover the incision and cannula.
- At this stage inject through the venous cannula 200 Units of heparin, dissolved in saline solution, per 100 g of body weight.
- Then tie in a carotid cannula of external diameter about 1 mm and connect by a column of saline solution containing heparin with a suitable pressure measuring device such as a mercury manometer of internal diameter about 2 to 3 mm.
- The central and peripheral nervous system including both vagus and associated sympathetic nerves is left intact. No artificial respiration is necessary.
- Taking care that no air is injected, inject all solutions through the venous cannula by means of a 1-ml syringe graduated in 0.01 ml and wash in with 0.2 ml of saline solution from the burette.
- Dilute the extract of the Standard Preparation and the preparation being examined with saline solution so that the volume to be injected is between 0.1 ml and 0.5 ml. Choose two doses of the Standard Preparation such that the elevation of the blood

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pressure is about 4 kPa for the lower dose and about 7 kPa but always submaximal for the higher dose, the ratio of low to high dose being determined by the response and usually being 3 to 5.

- As an initial approximation doses of 3 and 5 milliUnits may be tried. Choose two doses of the preparation being examined with the same inter-dose ratio, matching the effects of the dose of the Standard Preparation as closely as possible.
- Inject doses at intervals of 10 to 15 minutes. The two doses of the Standard Preparation and the two doses of the preparation being examined should be given in a randomised block or a Latin square design and four to five responses to each should be recorded.
- Measure all the responses and calculate the result of the assay by standard statistical methods.

BIOASSAY OF ACTH:

- ACTH (Adrenocorticotrophic hormone, corticotropin) is polypeptide tropic hormone (39 amino acids) secreted by the anterior pituitary gland.
- ACTH stimulates the production of cortisol, a steroid hormone important for regulating glucose, protein and lipid metabolism, suppressing the immune system response, and helping to maintain blood pressure.

Official Preparations

- Corticotropin injection: Is a sterile solution, in a suitable diluent, of the polypeptide from the pituitary glands of mammals. Potency range should be 80.0 – 120.0 % of USP corticotropin units.
- Corticotropin for injection, antimicrobial agent. Repository corticotropin injection is corticotropin in a sterile solution of partially hydrolyzed gelatin and is intended for subcutaneous and intramuscular use. This solution has been adopted as the reference standard for the bioassay.
 - **Packing:** Preserve in single-dose or multiple-dose containers of Type-1 glass.
 - **Storage:** Store in cold place.
 - **Labeling:** Injection recommends intravenous administration

Purpose and rationale

- This is a historical assay method. Administration of pituitary ACTH decrease the ascorbic acid present in the adrenals. The depletion of adrenal ascorbic acid is a function of the dose of ACTH administered. This relationship has been used for a quantitative assay of ACTH.

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Solutions:

- Solution A: Five units of test or standard dissolved in 0.25 ml of 0.5% phenol solution and diluted with 8.1 ml of 15% gelatin solution (Now 0.5ml contain 300 mU ACTH).
- solution B: Three ml of solution A diluted with 6 ml gelatin solution. Now concentration reduced to 100 mU ACTH/ 0.5 ml.
- solution C: Again 3 ml of solution B diluted with 6 ml of gelatin solution, the resulting solution contains 33 mU ACTH/ 0.5 ml.

Procedure

- Male Wistar rat (100-200 g) are hypophysectomized (pituitary gland removed by surgery) one day prior to the test.
- For one test with 3 dose of test preparation and standard solution used for the study.
- Number of hypophysectomized rats required: at least 36(preferably 60).
- The hypophysectomized rats are randomly distributed in to six groups. Each rat receives subcutaneous 0.5 ml of the various concentrations of test or standard.
- Three hours after injection, the animals are anesthetized and both adrenals removed, freed from extraneous tissue and weighed. The rats are sacrificed and the skull opened to verify completeness of hypophysectomy.
- The adrenals are homogenized in glass tubes contains 200 mg pure sand and 8.0 ml of 4% trichloroacetic acid and the ascorbic acid determined. (Roe and Kuether 1943).
- The potency ratio including confidence limits is calculated with the 3 + 3 point assay.

Ascorbic acid determination:

Reagents

- 0.02% ascorbic acid solution
- 85 % sulfuric acid (9N H₂SO₄)
- 0.02 g/ml of dinitrophenolhydrazine in 9N H₂SO₄
- 0.06 g/ml of thiourea are dissolved in distilled water
- Charcoal

Preparation of 0.02% ascorbic acid solution

- 100 mg L-ascorbic acid are dissolved in 100 ml of 4% trichloroacetic acid (1mg/ml solution) (Solution A= 1 % solution)
- 2 ml of Solution A diluted in 10 ml of 4% trichloroacetic acid to achieve a 0.2% ascorbic acid solution (solution B)

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- 1 ml of solution B diluted in 10 ml of 4% trichloroacetic acid to achieve a 0.02% ascorbic acid solution (solution C)

Preparation of other solutions

- Sulfuric acid (85%) is obtained by adding 900 ml concentrated sulfuric acid to 100 ml distilled water.
- Two g dinitrophenolhydrazine are dissolved in 100 ml 9 N H₂SO₄ (75 ml distilled water and 25 ml concentrated sulfuric acid).
- Six g thiourea are dissolved in 100 ml distilled water.

Calibration

- Trichloroacetic acid (4%) is added to 0.0, 0.5, 1.0, 2.0, 3.0, 4.0, 6.0, 8.0 ml of the 0.02% ascorbic acid solution (solution C) and 1.0, 1.5 and 2.0 ml of the 0.2% ascorbic acid solution to reach a final volume of 8.0 ml (Solution B).
- 100 mg charcoal is added to each sample and thoroughly mixed by shaking for 1 min.
- After 5 min the solutions are filtered.
- An aliquot of 0.1 ml of the 6% thiourea solution is added to 2.0 ml of the filtrate followed by 0.5 ml dinitrophenylhydrazine solution.
- The mixture is shaken and heated for 45 min at 57°C in a waterbath.
- The solutions are placed in an ice-cold water bath and with further cooling 2.5 ml of the 85% sulfuric acid are added.
- The calibration curve is established at a wave length of 540 nm using the solutions without ascorbic acid as blank.

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BIOASSAY OF HISTAMINE:

- Histamine is present in almost all the animal tissues mostly within mast cell and basophil granules. Tissues rich in histamine are skin, gastric and intestinal mucosa, lungs, liver and placenta.
- Non-mast cell histamine is present in brain, epidermis, gastric mucosa and growing regions. Histamine is also present in blood, most body secretions, venoms and pathological fluids. It is now known to play important physiological roles. Histamine

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produces effects by acting on the histamine receptors (H1, H2, H3 and H4) present on target cells.

- Bioassay of histamine on isolated guinea pig ileum can be determined by graded bioassay procedure i.e. i) Matching bioassay, ii) Interpolation bioassay, iii) Bracketing assay, iv) Multiple point assays.
- Bioassay of histamine in biological samples can be studied by using different bioassay methods. Depending upon pharmacological action, histamine can be assayed by:
 - Contractile effect of isolated ileum of guinea pig,
 - Contractile effect of uterus of guinea pig and
 - Fall in blood pressure of anaesthetized and atropinized cat. Guinea pig's uterus

Bioassay of histamine using guinea pig ileum:

Principle:

- Guinea pig ileum is very useful preparation for the bioassay methods. It is more sensitive to histamine. Contractile response of histamine to ileum is due to presence of H1 receptor.

Preparation of Standard and other solution:

- Prepare the stock of Tyrode solution. Also prepare the standard stock solution of histamine (1 mg/ml) and then different concentrations of histamine by serial dilution method.

Procedure:

- Sacrifice the 24 hr fasted guinea pig by stunning on the head and carotid bleeding. Fix the animal on the dissecting board by tying its legs. Open the abdominal cavity by a small horizontal cut followed by vertical midline incision and expose the abdominal organs.
- Trace the ileocaecal junction by lifting the caecum, then go upwards up to 8 cm from the junction and cut the 2-3 cm long segment of ileum muscle (excluding the terminal 5-8 cm, contains excess of excitatory α -adrenergic receptor, which may interfere in the study).
- Immediately place it in a petri dish containing aerated warm Tyrode solution. Slowly remove the mesentery attached to the muscle and then gently clean the lumen of ileum by passing luke warm Tyrode solution through it with a pipette or syringe.
- Mount the preparation in the inner organ bath containing Tyrode solution (20 ml) maintained at 37°C. Tie the bottom end of the muscle to the hook of aeration tube and the upper end to the isotonic frontal lever by a thread without closing the lumen.

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Adjust the magnification of response to 7-10 fold. Aerate the tissue with O₂ or carbogen slowly (40-60 bubbles per minutes).

- Stabilize the tissue for 30 min by applying a tension of 0.5 g weight attached to the lever, during which wash the tissue with fresh Tyrode solution once in every 10 min.
- Use 5 min time cycle with contact time of 30s for recording the contraction dependent responses of tissue due to histamine on the smoked drums.
- Record the responses of standard and test compounds.
- Record the responses of test compound i.e. unknown concentration of histamine with gradually increasing volume, till obtaining a response (T) which would lie on the linear portion (30-70%) of the CRC. Fix the response obtained due to volume of T.
- Record the graded responses of standard solution and test solution of histamine.
- The potency of the test sample is calculated in relation to that of the std. preparation by dividing the average lethal dose of the sample to the test and expressed as units per gram.

BIOASSAY OF 5-HT:

- 5-Hydroxytryptamine (5HT) is the biologically active local hormone, low molecular weight, and also an important neurotransmitter in the brain and periphery.
- It was first detected in serum (serotonin) and enterochromaffin cells of gut mucosa (enteramine), and later both were identified to be 5HT. Today the terms 5HT and serotonin are used interchangeably.
- About 90% of body's content of 5-HT is localized in the intestines; most of the rest is in platelets and brain. It is also found in wasp and scorpion sting, and is widely distributed in invertebrates and plants (banana, pear, pineapple, tomato etc.).
- The concentration of 5-HT in biological samples can be assayed by: Isolated fundus strip of rat stomach, Isolated terminal colon of rat, Isolated rat uterus, Perfused rabbit ear.

Isolated fundus strip of rat stomach

Principle:

- Rat fundus preparation is a very sensitive tissue for the bioassay methods and is useful to study the action of several substances like 5-HT, ACh, PGE₂ and bradykinin. Fundus strip preparation is slow contracting muscle. Fundus part of stomach can be easily identified by its grey colour and situated above the pink coloured thick pyloric part. A zig-zag preparation of the fundal strip is prepared so as to expose maximum portion of the tissue to drug.

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Preparation of Standard and other solution:

- Prepare the stock of Krebs solution. Prepare the standard stock solution of serotonin (1 mg/ml) and then different concentrations of serotonin by serial dilution method.

Procedure:

- Sacrifice the 24 hr fasted rat by stunning on the head and carotid bleeding. Fix the animal on the dissecting board by tying its legs. Open the abdominal cavity by a small horizontal cut followed by vertical midline incision and expose the abdominal organs.
- Identify the stomach and separate it from the abdomen by gently cutting its cardiac and pyloric end. Place it in a petri dish containing aerated warm Krebs solution.
- Incise the fundus of the stomach (upper grey part) from pyloric part (pink and thick part). Cut the fundus from the lesser curvature and open it longitudinally. Then cut the fundus at the midline into two equal parts. Give alternate transverse cuts (zig-zag cut) on opposite sides of the muscle to make a fundal strip preparation (as shown in fig. 9.1).
- Mount the preparation in the inner organ bath containing Krebs solution (20 ml) maintained at 37°C. Tie the bottom end of the muscle to the hook of aeration tube and the upper end to the isotonic frontal lever by thread. Adjust the magnification of response to 10-15 fold. Aerate the tissue with O₂ or carbogen slowly (40-60 bubbles per minutes).
- Apply 1 g load and stabilize the tissue for 30 min during which wash the tissue with fresh Krebs solution once in every 10 min.
- Use a 5 min time cycle with contact time of 90s (since the fundus strip muscle contracts slowly and relaxes slowly) for recording the contraction due to serotonin.
- Record the graded responses of standard solution and test solution of serotonin.
- The potency of the test sample is calculated in relation to that of the std. preparation by dividing the average lethal dose of the sample to the test and expressed as units per gram.

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